

THE RANK-TWO MARCUS PERMANENT INEQUALITY FOR REAL 3×3 MATRICES

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ABSTRACT. Marcus conjectured that

$$\operatorname{per} \begin{pmatrix} T & T \\ T & T \end{pmatrix} \leq \binom{2n}{n} \operatorname{per}(T)^2$$

for every real $n \times n$ matrix T of rank at most two. We prove the complete order-three case. The proof factors T through $\mathbb{R}^2$, separates the two projective multiplicity types of three nonzero row factors, and in each type reduces the gap to an explicit homogeneous sum of squares. We also classify equality: it holds precisely when $\operatorname{rank} T \leq 1$, when T has a zero row or column, or when a rank-two matrix can be permuted to contain a 2×2 all-zero submatrix. The decisive polynomial identities are accompanied by independent exact verifiers, one of which uses only sparse integer polynomial arithmetic and the definition of the permanent. This is a partial result toward the all-order conjecture: no assertion is made for $n \geq 4$ or for complex matrices.

1. INTRODUCTION

For an $m \times m$ matrix $A = (a_{ij})$, its permanent is

$$\operatorname{per}(A) = \sum_{\sigma \in S_m} \prod_{i=1}^m a_{i, \sigma(i)}.$$

Marcus proposed the following inequality for real matrices of rank at most two:

$$(1) \quad \operatorname{per} \begin{pmatrix} T & T \\ T & T \end{pmatrix} \leq \binom{2n}{n} \operatorname{per}(T)^2.$$

It appears as Conjecture 10 in his work on determinantal identities for rank-two permanents [1]. The same source reports that the positive-entry case is known, without citing a separate proof, and leaves the general signed case open.

The purpose of this note is to settle the first nontrivial small order beyond the immediate quadratic calculation. Our main result is the following.

Theorem 1.1. *Let $T \in \mathbb{R}^{3 \times 3}$ and $\operatorname{rank} T \leq 2$. Then*

$$(2) \quad \operatorname{per} \begin{pmatrix} T & T \\ T & T \end{pmatrix} \leq 20 \operatorname{per}(T)^2.$$

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Equality holds if and only if at least one of the following conditions holds:

- (1) $\text{rank } T \leq 1$;
- (2) T has a zero row or a zero column;
- (3) T has rank two and, after row and column permutations, contains a 2×2 all-zero submatrix.

The conditions are not required to be disjoint.

The proof is exact and does not divide by $\text{per}(T)$. This point matters: there are genuine rank-two equality matrices with vanishing permanent, for example

$$\begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 1 \\ 1 & 1 & 0 \end{pmatrix}.$$

We first quotient the internal $\text{GL}_2(\mathbb{R})$ freedom of a rank-two factorization, but retain homogeneous column coordinates so that directions at infinity are not lost. Three projective row directions have only two rank-two multiplicity types. The corresponding gaps are the two sums of squares in Lemma 3.1; their zero sets give the equality classification with no limiting argument.

Our literature search was frozen on August 29, 2026. Exact-title, formula, DOI, MathDB, and equivalent binary-form/apolar searches found no earlier order-three theorem or equality classification. This is a bounded search statement, not a claim about unindexed literature. The original all-order conjecture, and in particular the order-four cross-ratio problem, are not resolved here.

2. FACTORIZATION, GAUGE, AND HOMOGENEITY

For $T \in \mathbb{R}^{3 \times 3}$ of rank at most two, choose

$$(3) \quad T = UV^T, \quad U, V \in \mathbb{R}^{3 \times 2}.$$

Write the rows of U and V as u_i and v_j , so that $T_{ij} = u_i v_j^T$. For every $G \in \text{GL}_2(\mathbb{R})$, the change

$$(4) \quad (U, V) \mapsto (UG, VG^{-T})$$

preserves T . Thus the nonzero rows of U and V define projective points on which the usual real projective action is available.

Define the order-three gap

$$(5) \quad \mathcal{D}(T) := 20 \text{per}(T)^2 - \text{per} \begin{pmatrix} T & T \\ T & T \end{pmatrix}.$$

If $D_r = \text{diag}(r_1, r_2, r_3)$ and $D_s = \text{diag}(s_1, s_2, s_3)$, direct homogeneity gives

$$(6) \quad \mathcal{D}(D_r T D_s) = \left(\prod_{i=1}^3 r_i \prod_{j=1}^3 s_j \right)^2 \mathcal{D}(T).$$

We use this identity only with nonzero scales in normalizations. A zero scale creates a zero row or column and will be handled directly.

Suppose for the moment that T has rank two and no zero row or column. We may choose both factor matrices in (3) to have rank two; then all u_i, v_j are nonzero. The three row directions $[u_i] \in \mathbb{RP}^1$ are either all distinct or have multiplicity type $2 + 1$. Three coincident directions would force rank $U = 1$.

In the distinct case, projective triple transitivity, (4), and nonzero row scalings put the matrix into the form

$$(7) \quad A_{ij} = c_j + a_i d_j, \quad (a_1, a_2, a_3) = (-1, 0, 1),$$

where each homogeneous column pair (c_j, d_j) is nonzero. In the $2 + 1$ case the corresponding form is

$$(8) \quad A_{ij} = c_j + a_i d_j, \quad (a_1, a_2, a_3) = (0, 0, 1).$$

No division by a column coordinate has occurred, so both forms include projective column directions at infinity.

3. THE TWO EXACT SUMS OF SQUARES

For homogeneous column pairs, set

$$(9) \quad [ij] := d_i c_j - c_i d_j.$$

Lemma 3.1 (Normal-form identities). *For the distinct form (7),*

$$(10) \quad \mathcal{D}(A) = 16([23]^2(d_1^2 + 3c_1^2) + [13]^2(d_2^2 + 3c_2^2) + [12]^2(d_3^2 + 3c_3^2)).$$

For the repeated form (8),

$$(11) \quad \mathcal{D}(A) = 16([12]^2 c_3^2 + [13]^2 c_2^2 + [23]^2 c_1^2).$$

Proof. Both statements are polynomial identities with integer coefficients. They follow by substituting (7) or (8) into

$$20 \left(\sum_{\sigma \in S_3} \prod_i A_{i, \sigma(i)} \right)^2 - \sum_{\tau \in S_6} \prod_{i=1}^6 A_{\bar{i}, \tau(\bar{i})},$$

where $\bar{i}$ denotes reduction modulo three into $\{1, 2, 3\}$. Collecting monomials yields (10) and (11). This is a finite direct derivation from the permanent definition.

For independent reproducibility, the supplied standard-library verifier represents a polynomial by a sparse dictionary from six-variable exponent tuples to integer coefficients, enumerates all $3!$ and $6!$ permutations, constructs the right-hand sides independently, and checks that both remainders have no monomials. A separate SymPy verifier performs the same reconstruction through a different polynomial implementation. $\square$

4. PROOF AND EQUALITY CASES

Proof of Theorem 1.1. First assume that T has rank two and no zero row or column. By the previous sections and (6), it suffices to work with one of the two normal forms.

If the row directions are distinct, every weight $d_j^2 + 3c_j^2$ in (10) is strictly positive. Hence the gap is nonnegative. Equality would force

$$[12] = [13] = [23] = 0,$$

so all column directions would coincide and V would have rank one. This contradicts $\text{rank } T = 2$. Therefore the inequality is strict on this stratum.

Now suppose that the row directions have type $2 + 1$. The second sum of squares proves nonnegativity. We determine its zero set under the standing assumptions that all column pairs are nonzero and V has rank two. If all c_j are nonzero, equality forces all brackets to vanish, again contradicting $\text{rank } V = 2$. If exactly one c_j vanishes, say $c_3 = 0$, then $d_3 \neq 0$ and

$$[13]^2 c_2^2 = c_1^2 d_3^2 c_2^2 > 0.$$

All three c_j cannot vanish because V has rank two. Thus equality occurs exactly when, after a column permutation,

$$(12) \quad c_1 = c_2 = 0, \quad c_3 \neq 0.$$

In (8), this says precisely that the two proportional rows and the first two columns form a 2×2 zero submatrix.

Conversely, if such a zero block is placed in the upper left, then

$$T = \begin{pmatrix} 0 & 0 & a \\ 0 & 0 & b \\ c & d & e \end{pmatrix}.$$

The first two rows cannot be matched to two distinct columns, so $\text{per}(T) = 0$. In the doubled matrix, the four copies of those rows can use at most the two copies of the third column, so the block permanent also vanishes.

If T has a zero row or column, both permanents vanish directly. Finally, if $\text{rank } T \leq 1$, write $T = xy^\top$. Then

$$\text{per}(T) = 3! \prod_i x_i \prod_j y_j$$

and

$$\text{per} \begin{pmatrix} T & T \\ T & T \end{pmatrix} = 6! \left(\prod_i x_i \prod_j y_j \right)^2 = 20 \text{per}(T)^2.$$

These cases exhaust rank at most two and establish the result. $\square$

5. BINARY-FORM REDUCTION AND AN INDEPENDENT PROOF ROUTE

There is a useful invariant reformulation that independently checks the normal-form calculation. From (3), form the split real binary cubics

$$f(X, Y) = \prod_{i=1}^3 (u_{i0}X + u_{i1}Y) = \sum_{k=0}^3 f_k X^{3-k} Y^k$$

and define g similarly from the v_j . For degree- m forms put

$$[p, q]_m := \sum_{k=0}^m \frac{p_k q_k}{\binom{m}{k}}.$$

Grouping permanent terms according to the number of second-coordinate choices gives

$$(13) \quad \text{per}(T) = 3![f, g]_3, \quad \text{per}\begin{pmatrix} T & T \\ T & T \end{pmatrix} = 6![f^2, g^2]_6.$$

Since $20(3!)^2 = 6!$, the theorem is equivalent to

$$(14) \quad [f^2, g^2]_6 \leq [f, g]_3^2$$

for split real binary cubics.

On the three-distinct-factor stratum, a dual projective change sends f to $XY(X + Y)$. Writing the three ordered affine roots of g as $t, t + p, t + p + q$ with $p, q \geq 0$ turns the gap into a quadratic in t . Its leading coefficient is $2(p^2 + pq + q^2)$ and its discriminant is

$$-12(p^4 q^2 + p^4 + 2p^3 q^3 + 2p^3 q + p^2 q^4 + 3p^2 q^2 + 2pq^3 + q^4),$$

which is negative unless $p = q = 0$. The repeated-factor stratum reduces in a finite chart to

$$g_1^2 - 3g_0 g_2 = \frac{g_0^2}{2} \sum_{i < j} (r_i - r_j)^2.$$

The independent builder and referee reports treat the homogeneous boundary cases explicitly. This also shows why split real cubics are essential: the coefficient quadratic form is not nonnegative on arbitrary cubics.

For degree four, the same reduction leaves a genuine cross-ratio parameter after three row directions are normalized. Controlling that parameter and all collision strata is unresolved; neither this cubic proof nor the recorded numerical searches certify the $n = 4$ case.

6. EXACT VERIFICATION AND DATA AVAILABILITY

The exact certificate and all verification code accompany this note. From the project root, the core checks are

```
python certificates/verify_n3_sos_stdlib.py
python certificates/verify_n3_sos.py
python certificates/test_fail_closed.py
bash scripts/reproduce_clean.sh
```

The first verifier uses only the Python standard library and parses the serialized file

`certificates/n3_sos_certificate.json`.

It reconstructs both permanents from the definition and fails on any nonzero remainder. The second uses SymPy through a separate implementation. Both bind the canonical certificate hash and enforce an exact nested JSON schema. The negative suite requires them to reject byte-hash, extra-key, dropped-key, and data-tampering cases. The clean runner enforces CPython 3.13.5, installs SymPy 1.13.3 from a hash-locked requirements file in a fresh virtual environment, runs the equality, discriminant, referee, and breaker checks, and builds the paper from clean sources. The recorded integer searches for $n = 3$ and $n = 4$ are diagnostics only and are not used in the proof.

The literature claim ledger, exact search queries, route registry, proof gap ledger, adversarial reports, and release hashes are included in the project archive. A result-specific second search, including matrix, binary-cubic, apolar, citation, arXiv, Crossref, and MathDB queries, found no prior occurrence in the dated databases and queries recorded there. This is not a claim of global novelty.

COMPUTATIONAL AND AI DISCLOSURE

OpenAI Codex was used to assist literature discovery, counterexample search, symbolic expansion, proof auditing, verifier implementation, and manuscript drafting. Three separated roles were used for proof construction, counterexample hunting, and independent referee reconstruction. Every decisive algebraic identity is exact and is rebuilt by independent programs from serialized input or from the permanent definition. Floating-point optimization was used only for falsification attempts and has no role in the theorem. No proof assistant, SMT solver, or computer algebra positivity oracle was used to establish the final nonnegativity: after the exact identities, it is the displayed sum-of-squares argument.

REFERENCES

1. Adam W. Marcus, *A determinantal identity for the permanent of a rank 2 matrix*, The American Mathematical Monthly **129** (2022), no. 10, 962–971.

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